

Name :-

Javeria Hamid

SAP ID :-

70148901

Section :-

(0)

Submitted To :-

Dr. Nadeem

Assignment No 3 :-

Exercise 2.4 :-

1 to 8, 25, 26, 29 to 34, 39, 40

Question no 1 :-

Find these terms of the sequence $\{a_n\}$,
where $a_n = 2 \cdot (-3)^n + 5^n$

(a)

a_0

$$\begin{aligned} a_n &= 2 \cdot (-3)^n + 5^n \\ &= 2 \cdot (-3)^0 + 5^0 \\ &= 2(1) + (1) \\ &= 2 + 1 \\ &= \boxed{3} \end{aligned}$$

(b)

a_1

$$\begin{aligned} a_n &= 2 \cdot (-3)^n + 5^n \\ &= 2 \cdot (-3)^1 + 5^1 \\ &= 2 \cdot (-3) + 5 \\ &= -6 + 5 \\ &= \boxed{-1} \end{aligned}$$

(c)

a_4

$$\begin{aligned} a_n &= 2 \cdot (-3)^n + 5^n \\ &= 2 \cdot (-3)^4 + 5^4 \\ &= 2(81) + 625 \\ &= 162 + 625 \\ &= \boxed{787} \end{aligned}$$

(d)

a^5

$$a_n = 2 \cdot (-3)^n + 5^n$$

$$= 2 \cdot (-3)^5 + 5^5$$

$$= 2 \cdot (-243) + 3125$$

$$= -486 + 3125$$

$$= \boxed{2639}$$

Question 10 :-

What is the term a_8 of the sequence $\{a_n\}$ if a_n equals.

(a)

$$2^n - 1$$

$$= 2^8 - 1$$

$$= 27$$

$$= \boxed{128}$$

(b)

$$7^n$$

$$a^8 = 7$$

(c)

$$1 + (-1)^n$$

$$= 1 + (-1)^8$$

$$= 1 + 1$$

$$= \boxed{2}$$

(d)

$$-(-2)^n?$$

$$= -(-2)^8$$

$$= -(256)$$

$$= \boxed{-256}$$

Question no 3:-

What are the terms a_0, a_1, a_2 and a_3 of the sequence $\{a_n\}$, where a_n equals:

(a)

$$2^n + 1?$$

$$a_0 = 2^0 + 1 = (1) + 1 = \boxed{2}$$

$$a_1 = 2^1 + 1 = 2 + 1 = \boxed{3}$$

$$a_2 = 2^2 + 1 = 4 + 1 = \boxed{5}$$

$$a_3 = 2^3 + 1 = 8 + 1 = \boxed{9}$$

(b)

$$(n+1)^{n+1}?$$

$$a_0 = (0+1)^{0+1} = (1)^1 = \boxed{1}$$

$$a_1 = (1+1)^{1+1} = (2)^2 = \boxed{4}$$

$$a_2 = (2+1)^{2+1} = (3)^3 = \boxed{27}$$

$$a_3 = (3+1)^{3+1} = (4)^4 = \boxed{256}$$

(c)

$$\lfloor n/2 \rfloor?$$

$$a_0 = \lfloor 0/2 \rfloor = \lfloor 0 \rfloor = 0$$

$$a_1 = \lfloor 1/2 \rfloor = \lfloor 0.5 \rfloor = 0$$

$$a_2 = \lfloor 2/2 \rfloor = \lfloor 1 \rfloor = 1$$

$$a_3 = \lfloor 3/2 \rfloor = \lfloor 1.5 \rfloor = 1$$

Point Values: less and Greater.

(d)

$$\lfloor n/2 \rfloor + \lceil n/2 \rceil ?$$

$$a_0 = \lfloor 0/2 \rfloor + \lceil 0/2 \rceil = \lfloor 0 \rfloor + \lceil 0 \rceil = 0 + 0 = \boxed{0}$$

$$a_1 = \lfloor 1/2 \rfloor + \lceil 1/2 \rceil = \lfloor 0.5 \rfloor + \lceil 0.5 \rceil = 0 + 1 = \boxed{1}$$

$$a_2 = \lfloor 2/2 \rfloor + \lceil 2/2 \rceil = \lfloor 1 \rfloor + \lceil 1 \rceil = 1 + 1 = \boxed{2}$$

$$a_3 = \lfloor 3/2 \rfloor + \lceil 3/2 \rceil = \lfloor 1.5 \rfloor + \lceil 1.5 \rceil = 1 + 2 = \boxed{3}$$

Question No 4.

What are the terms a_0, a_1, a_2 and a_3 of the sequence $\{a_n\}$, where a_n equals.

(a)

$$(-2)^n ?$$

$$a_0 = (-2)^0 = \boxed{1}$$

$$a_1 = (-2)^1 = \boxed{-2}$$

$$a_2 = (-2)^2 = \boxed{4}$$

$$a_3 = (-2)^3 = \boxed{-8}$$

(b)

3?

$$a_0 = 3$$

$$a_1 = 3$$

$$a_2 = 3$$

$$a_3 = 3$$

(c)

$$7 + 4^n ?$$

$$a_0 = 7 + 4^0 = 7 + 1 = \boxed{8}$$

$$a_1 = 7 + 4^1 = 7 + 4 = \boxed{11}$$

$$a_2 = 7 + 4^2 = 7 + 16 = \boxed{23}$$

$$a_3 = 7 + 4^3 = 7 + 64 = \boxed{71}$$

(d)

$$2^n + (-2)^n?$$

$$a_0 = 2^0 + (-2)^0 = 1 + (1) = \boxed{2}$$

$$a_1 = 2^1 + (-2)^1 = 2 + (-2) = \boxed{0}$$

$$a_2 = 2^2 + (-2)^2 = 4 + (4) = \boxed{8}$$

$$a_3 = 2^3 + (-2)^3 = 8 + (-8) = \boxed{0}$$

Successive term: Jo ans aata hai preceding answer ka picla wali

Question is 10 5 :-

List the first 10 terms of each of these sequences.

(a)

the sequence that begins with $\boxed{2}$ and in which each successive term is 3 more than the preceding term.

3 times preceding terms.

2, 5, 8, 11, 14, 17, 20, 23, 26, 29

(b)

the sequence that lists each positive integer three times, in increasing order.

1, 1, 1, 2, 2, 2, 3, 3, 3, 4

(c)

the Sequence that lists the odd positive integers in increasing order, listing each odd integer twice.

1, 1, 3, 3, 5, 5, 7, 7, 9, 9.

(d)

the Sequence whose n th term is $n! - 2^n$.

$$1! - 2^1 = 1 - 2 = -1$$

$$2! - 2^2 = 2 - 4 = -2$$

$$3! - 2^3 = 6 - 8 = -2$$

$$4! - 2^4 = 24 - 16 = 8$$

$$5! - 2^5 = 120 - 32 = 88$$

$$6! - 2^6 = 720 - 64 = 656$$

$$7! - 2^7 = 5040 - 128 = 4912$$

$$8! - 2^8 = 40320 - 256 = 40064$$

$$9! - 2^9 = 362880 - 512 = 362368$$

(e)

the Sequence that begins with 3, where each succeeding term is twice the preceding term.

$$3 \times 2 = 6$$

$$6 \times 2 = 12$$

$$12 \times 2 = 24$$

$$24 \times 2 = 48$$

$$48 \times 2 = 96$$

$$96 \times 2 = 192$$

$$192 \times 2 = 384$$

$$384 \times 2 = 768$$

$2 \times 2 \times 2$

$$1536 \times 2 = 3072 \text{ (d)}$$

$$1536 \times 2 = 3072 \text{ (d)}$$

the sequence whose first term is 2, second term is 4, and each succeeding term is the sum of the two preceding terms.

$$2 + 4 = 6$$

$$10 + 6 = 16$$

$$\underline{26 + 16 = 42}$$

$$68 + 42 = 110$$

$$178 + 110 = 288$$

$$288 + 178 = 466$$

the sequence whose n th term is the number of bits in the binary expansion of the number n

1.1, 2.2, 3.3, 3.3, 4.4, 4.4

$$\begin{array}{r} 2 \overline{) 210} \\ \underline{2} \\ 0 \end{array}$$

10 → 1111

$$\begin{array}{r} 3 \\ 1 - 6 \end{array}$$

$$\begin{array}{r} 2 \overline{) 15} \\ \underline{14} \\ 1 \end{array}$$

(h)

the sequence where the n th term is the number of letters in the English word from the index n .

One = 3

Two = 3

Three = 5

Four = 4

Five = 4

Six = 3

Seven = 5

Eight = 5

Nine = 4

Ten = 3

Question No 6 :-

List the first 10 terms of each of these sequences.

(a)

the sequence obtained by starting with 10 and obtaining each term by subtracting 3 from the previous term:

$$a_1 = 10$$

$$a_2 = 10 - 3 = 7$$

$$a_3 = 7 - 3 = 4$$

$$a_4 = 4 - 3 = 1$$

$$a_5 = 1 - 3 = -2$$

$$a_6 = -2 - 3 = -5$$

$$a_7 = -5 - 3 = -8$$

$$a_8 = -8 - 3 = -11$$

$$a_9 = -11 - 3 = -14$$

$$a_{10} = -14 - 3 = -17$$

(b)

the sequence whose n th term is the sum of the n positive integer.

Formula: for positive integer

$$S_n = \frac{n(n+1)}{2}$$

$$S_1 = \frac{1(1+1)}{2} = \frac{1(2)}{2} = 1$$

$$S_2 = \frac{2(2+1)}{2} = \frac{2(3)}{2} = 3$$

$$S_3 = \frac{3(3+1)}{2} = \frac{3(4)}{2} = 6$$

$$S_4 = \frac{4(4+1)}{2} = \frac{4(5)}{2} = 10$$

$$S_5 = \frac{5(5+1)}{2} = \frac{5(6)}{2} = 15$$

$$S_6 = \frac{6(6+1)}{2} = \frac{6(7)}{2} = 21$$

$$S_7 = \frac{7(7+1)}{2} = \frac{7(8)}{2} = 28$$

$$S_8 = \frac{8(8+1)}{2} = \frac{8(9)}{2} = 36$$

$$S_9 = \frac{9(9+1)}{2} = \frac{9(10)}{2} = 45$$

$$S_{10} = \frac{10(10+1)}{2} = \frac{10(11)}{2} = 55$$

(C)

the sequence whose n^{th} term is $3^n - 2^n$

$$a_1 = 3^{(1)} - 2^{(1)} \\ = 3 - 2 = \boxed{1}$$

$$a_2 = 3^{(2)} - 2^{(2)} = 9 - 4 = 5$$

$$a_3 = 3^{(3)} - 2^{(3)} = 27 - 8 = 19$$

$$a_4 = 3^4 - 2^4 = 81 - 16 = 65$$

$$a_5 = 3^5 - 2^5 = 243 - 32 = 211$$

$$a_6 = 3^6 - 2^6 = 729 - 64 = 665$$

$$a_7 = 3^7 - 2^7 = 2187 - 128 = 2059$$

$$a_8 = 3^8 - 2^8 = 6561 - 256 = 6305$$

$$a_9 = 3^9 - 2^9 = 19683 - 512 = 19171$$

$$a_{10} = 3^{10} - 2^{10} = 59049 - 1024 = 58025$$

(d)

the sequence whose n^{th} term is $\lfloor \sqrt{n} \rfloor$

$$a_1 = \sqrt{1} = \lfloor 1 \rfloor = 1$$

$$a_2 = \sqrt{2} = \lfloor 1.4 \rfloor = 1$$

$$a_3 = \sqrt{3} = \lfloor 1.7 \rfloor = 1$$

$$a_4 = \sqrt{4} = \lfloor 2 \rfloor = 2$$

$$a_5 = \sqrt{5} = \lfloor 2.2 \rfloor = 2$$

$$a_6 = \sqrt{6} = \lfloor 2.4 \rfloor = 2$$

$$a_7 = \sqrt{7} = \lfloor 2.6 \rfloor = 2$$

$$a_8 = \sqrt{8} = \lfloor 2.8 \rfloor = 2$$

$$a_9 = \sqrt{9} = \lfloor 3 \rfloor = 3$$

$$a_{10} = \sqrt{10} = \lfloor 3.1 \rfloor = 3$$

(e)

the sequence whose first two terms are 1 and 5 and each succeeding term is the sum of the two previous terms.

1, 5, 6, 11, 17

$\boxed{1, 5}$

$$1 + 5 = 6$$

$$6 + 5 = 11$$

$$11 + 6 = 17$$

$$17 + 11 = 28$$

$$28 + 17 = 45$$

$$45 + 28 = 73$$

$$73 + 45 = 118$$

$$118 + 73 = 191$$

$$191 + 118 = 309$$

$$309 + 191 = 500$$

(f)

the sequence whose n th term is the largest integer whose binary expansion has n bits (write your ans in decimal notation).

Largest integers with n bits is $2^n - 1$

$$\boxed{2^n - 1} = 2^1 - 1 = 2 - 1 = \boxed{1}$$

$$2^2 - 1 = 4 - 1 = \boxed{3}$$

$$2^3 - 1 = 8 - 1 = \boxed{7}$$

$$2^4 - 1 = 16 - 1 = \boxed{15}$$

$$2^5 - 1 = 32 - 1 = \boxed{31}$$

$$2^6 - 1 = 64 - 1 = \boxed{63}$$

$$2^7 - 1 = 128 - 1 = \boxed{127}$$

$$2^8 - 1 = 256 - 1 = \boxed{255}$$

$$2^9 - 1 = 512 - 1 = \boxed{511}$$

$$2^{10} - 1 = 1024 - 1 = \boxed{1023}$$

(g)

the Sequence whose terms are constructed sequentially as follows: Start with 1, then add 1, then multiply by 1, then add 2, then multiply by 2, and so on.

Row

Add 1:- 1

$$1 + 1 = \boxed{2}$$

$$2 \times 1 = \boxed{2}$$

Add 2:- $2 + 2 = \boxed{4}$

$$4 \times 2 = \boxed{8}$$

Add 3:- $8 + 3 = \boxed{11}$

$$11 \times 3 = \boxed{33}$$

Add 4:- $33 + 4 = \boxed{37}$

$$37 \times 4 = \boxed{148}$$

Add 5:- $148 + 5 = \boxed{153}$

$$153 \times 5 = \boxed{765}$$

$$\begin{array}{l} 1 + 1 = 2 \\ 2 \times 1 = 2 \\ 2 + 2 = 4 \\ 4 \times 2 = 8 \\ 8 + 3 = 11 \\ 11 \times 3 = 33 \\ 33 + 4 = 37 \\ 37 \times 4 = 148 \\ 148 + 5 = 153 \\ 153 \times 5 = 765 \end{array}$$

1, 2, 2, 4, 8, 11, 33, 37, 148, 153, 765

(h)

the Sequence whose n th term is the largest integer k such that $k! \leq n$

$$1! \leq 1$$

$$1! = \boxed{1} \text{ So } 1! \leq 1$$

$$2! \leq 2 \quad 2 \leq 2$$

$$3! = 6 \quad 2! \leq 3$$

$$\textcircled{2} \quad 2! = 2$$

$$2! \leq 2$$

$$\textcircled{3} \quad 3! = 3 \times 2 \times 1 = 6$$

$$6 \leq 3$$

$$2! \leq 3$$

$$\textcircled{4} \quad 4! = 4 \times 3 \times 2 \times 1 = 24$$

$$24 \leq 4$$

$$2! \leq 4$$

$$\textcircled{5} \quad 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

$$120 \leq 5$$

$$2! \leq 5$$

$$\textcircled{6} \quad 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

$$720 \leq 6$$

$$3! \leq 6$$

$$\textcircled{7} \quad 7! = 5040$$

$$5040 \leq 7$$

$$3! \leq 7$$

$$\textcircled{8} \quad 8! = 40320$$

$$40320 \leq 8$$

$$3! \leq 8$$

$$\textcircled{9} \quad 9! = 362880$$

$$362880 \leq 9$$

$$3! \leq 9$$

$$\textcircled{10} \quad 10! = 3628800$$

$$3628800 \leq 10$$

$$3! \leq 10$$

1, 2, 2, 2, 2, 3, 3, 3, 3

Question No 7:-

Find at least three different sequences beginning with the terms 1, 2, 4 whose terms are generated by a simple formula or rule.

Formula:-

$2 \times 2 \times 2 \times 2$ $a_n = 2^n$

$$a_0 = 2^0 = 1$$

$$a_1 = 2^1 = 2$$

$$a_2 = 2^2 = 4$$

$$a_3 = 2^3 = 8$$

$$a_4 = 2^4 = 16$$

$$a_5 = 2^5 = 32$$

$$a_6 = 2^6 = 64$$

$$a_7 = 2^7 = 128$$

$$a_8 = 2^8 = 256$$

$$a_9 = 2^9 = 512$$

$$a_{10} = 2^{10} = 1024$$

Question No 8:-

Find at least three different sequences beginning with the terms 3, 5, 7, whose terms are generated by a simple formula or rule.

Formula :- $a_n = 3 + 2n$

$$a_0 = 3 + 2(0) = 3 + 0 = \boxed{3}$$

$$a_1 = 3 + 2(1) = 3 + 2 = \boxed{5}$$

$$a_2 = 3 + 2(2) = 3 + 4 = \boxed{7}$$

$$a_3 = 3 + 2(3) = 3 + 6 = \boxed{9}$$

$$a_4 = 3 + 2(4) = 3 + 8 = \boxed{11}$$

$$a_5 = 3 + 2(5) = 3 + 10 = \boxed{13}$$

$$a_6 = 3 + 2(6) = 3 + 12 = \boxed{15}$$

$$a_7 = 3 + 2(7) = 3 + 14 = \boxed{17}$$

$$a_8 = 3 + 2(8) = 3 + 16 = \boxed{19}$$

$$a_9 = 3 + 2(9) = 3 + 18 = \boxed{21}$$

$$a_{10} = 3 + 2(10) = 3 + 20 = \boxed{23}$$

= 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23

Three terms of the Sequence.

Question No. 25 :-

(a)
 $\overset{1}{1}, \overset{2}{0}, \overset{2}{1}, \overset{2}{0}, \overset{3}{1}, \overset{3}{1}, \overset{3}{0}, \overset{3}{0}, \overset{3}{1}, \dots$

1, 1, 1, 1

(b)
 $\overset{1}{1}, \overset{2}{2}, \overset{2}{2}, \overset{3}{3}, \overset{4}{4}, \overset{4}{4}, \overset{5}{5}, \overset{6}{6}, \overset{6}{6}, \overset{7}{7}, \overset{8}{8}, \overset{8}{8}, \dots$

9, 10, 10

(c)

$$1, 0, 2, 0, 4, 0, 8, 0, 16, 0, \dots$$

$$16 \times 2 = 32, \quad 32 \times 2 = 64$$

$$32, 0, 64$$

(d)

$$3, 6, 12, 24, 48, 96, 192, \dots$$

$$192 \times 2 = 384$$

$$384 \times 2 = 768$$

$$768 \times 2 = 1536$$

$$384, 768, 1536$$

27+7

(e)

$$15, 8, 1, -6, -13, -20, -27, \dots$$

$$- (7)$$

$$= -34$$

$$34 + 7 = 41$$

$$= -41$$

$$41 + 7 = 48$$

$$= \boxed{-48}$$

$$-34, -41, -48$$

(f)

$$3, 5, 8, 12, 17, 23, 30, 38, 47, \dots$$

$$3 + 2 = 5$$

$$5 + 3 = 8$$

$$8 + 4 = 12$$

$$12 + 5 = 17$$

$$17 + 6 = 23$$

$$23 + 7 = 30$$

$$30 + 8 = 38$$

$$38 + 9 = 47$$

$$47 + 10 = 57$$

$$57 + 11 = 68$$

$$68 + 12 = 80$$

$$= 57, 68, 80$$

$$2, 16, 54, 128, 250, 432, 686, \dots$$

Formula:- $a_n = 2n^3$

$$a_2 = 2(2)^3 = 2(8) = 16$$

$$a_6 = 2(6)^3 = 432$$

$$2(7)^3 = 2(343) = 686$$

$$2(8)^3 = 2(512) = 1024$$

$$2(9)^3 = 2(729) = 1458$$

$$2(10)^3 = 2(1000) = 2000$$

$$= 1024, 1458, 2000$$

$$2, 3, 7, 25, 121, 721, 5041, 40321, \dots$$

Formula:- $a_n = a_{n-1} \times (n-1) + 1$

$$a_9 = a_{9-1} \times (9-1) + 1$$

$$= a_8 \times (8) + 1$$

$$= 40321 \times 8 + 1$$

$$= 282247$$

$$= \boxed{322569}$$

$$a_{10} = a_{10-1} \times (10-1) + 1$$

$$= a_9 \times (9) + 1$$

$$= 322569 \times 9 + 1$$

$$= 2903121 + 1$$

$$= \boxed{2903122}$$

$$a_{11} = a_{11-1} \times (11-1) + 1$$

$$= a_{10} \times (10) + 1$$

$$= 2903122 \times 10 + 1$$

$$= 29031220 + 1$$

$$= \boxed{29031221}$$

$$322569, 2903122, 29031221, \dots$$

Question 10: 26: -

$$\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 3, & 6, & 11, & 18, & 27, & 38, & 51, & 66, & 83, & 102, \dots \end{matrix}$$

Formula :- $a_n = n^2 + 2$

$$a_1 = (1)^2 + 2$$

$$= 1 + 2 = \boxed{3}$$

$$a_{10} = (10)^2 + 2$$

$$= 100 + 2 = \boxed{102}$$

$$a_{11} = (11)^2 + 2$$

$$= 121 + 2 = \boxed{123}$$

$$a_{12} = (12)^2 + 2$$

$$= 144 + 2 = \boxed{146}$$

$$a_{13} = (13)^2 + 2$$

$$= 169 + 2 = \boxed{171}$$

$$= 123, 146, 171$$

(b)

$$7, 11, 15, 19, 23, 27, 31, 35, 39, 43, \dots$$

Arithmetic Seq :- $a_n = 7 + (n-1) \cdot 4$

$$a_1 = 7 + (1-1) \cdot 4$$

$$a_1 = 7 + (0) \cdot 4$$

$$= \boxed{7}$$

$$a_{11} = 7 + (11-1) \cdot 4$$

$$= 7 + (10) \cdot 4$$

$$= 7 + 40$$

$$= \boxed{47}$$

$$a_{12} = 7 + (12-1) \cdot 4$$

$$= 7 + (11) \cdot 4$$

$$= 7 + 44$$

$$= \boxed{51}$$

$$a_{13} = 7 + (13-1) \cdot 4$$

$$= 7 + (12) \cdot 4$$

$$= 7 + 48$$

$$= \boxed{55}$$

$$= 47, 51, 55$$

(c)
 $1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, \dots$

12 in binary = 1100

13 = 1101

14 = 1110

(d)
 $1, \underline{2}, \underline{2}, \underline{2}, \underline{3}, \underline{3}, \underline{3}, \underline{3}, \underline{3}, \underline{5}, \underline{5}, \underline{5}, \underline{5}, \underline{5}, \underline{5}, \underline{5}, \dots$
 $8, 8, 8$

(e)
 $0, 2, 8, 26, 80, 242, 728, 2186, 6560, 19682, \dots$

Formula = $3^n - 1$

$$a_{12} = 3^{12} - 1$$

$$= 531441 - 1$$

$$= 531440$$

$$a_{11} = 3^{11} - 1$$

$$= 177147 - 1$$

$$= 177146$$

$$a_{13} = 3^{13} - 1$$

$$= 1594323 - 1$$

$$= 1594322$$

(f)

1, 3, 15, 105, 945, 10395, 135135,
 2027025, 34459425, ...

Factorial pattern:

Multiply 2 numbers (odd)

$$(2n-1)!!$$

$$1 \times 3 = 3$$

$$3 \times 5 = 15$$

$$15 \times 7 = 105$$

$$105 \times 9 = 945$$

$$945 \times 11 = 10395$$

$$10395 \times 13 = 135135$$

$$135135 \times 15 = 2027025$$

$$2027025 \times 17 =$$

$$34459425$$

$$34459425 \times 19 = 654729075$$

$$654729075 \times 21 = 13746706075$$

$$13746706075 \times 23 = 316146289725$$

$$316146289725 \times 25 = 7903657243125$$

(g)

1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1, ...

0 repeat 6 times so the next three
 terms are 0, 0, 0.

(h)

2, 4, 16, 256, 65536, 4294967296, ...

$$a_n = 4^{n-1}$$

$$a_7 = 4^{7-1}$$

$$= 4^6$$

$$= 4096$$

$$a_8 = 4^{n-1}$$

$$= 4^{8-1}$$

$$= 4^7$$

$$= \boxed{16384}$$

$$a_9 = 4^{n-1}$$

$$= 4^{9-1}$$

$$= 4^8$$

$$= \boxed{65536}$$

$$= 4096, 16384, 65536$$

Question No 29 :-

What are the values of these sums?

(a)

$$\sum_{k=1}^5 (k+1)$$

Sol :-

$$(1+1) = 2$$

$$(2+1) = 3$$

$$(3+1) = 4$$

$$(4+1) = 5$$

$$(5+1) = 6$$

$$= 2 + 3 + 4 + 5 + 6 = \boxed{20}$$

(b)

$$\sum_{j=0}^4 (-2)^j$$

0 to 4

$$(-2)^0 = \boxed{1}$$

$$(-2)^1 = \boxed{-2}$$

$$(-2)^2 = \boxed{4}$$

$$(-2)^3 = \boxed{-8}$$

$$(-2)^4 = \boxed{16}$$

$$1 - 2 + 4 - 8 + 16 = \boxed{11}$$

(c)

$$\sum_{i=1}^{10} 3$$

$$3 \times 10 =$$

$$\boxed{30}$$

(d)

$$\sum_{j=0}^8 (2^{j+1} - 2^j)$$

$$0 - 8$$

$$= (2^{0+1} - 2^0) = (2^1 - 2^0) = (2 - 1) = 1$$

$$= (2^{1+1} - 2^1) = (2^2 - 2) = 2$$

$$= (2^{2+1} - 2^2) = (2^3 - 2^2) = 4$$

$$= (2^{3+1} - 2^3) = (2^4 - 2^3) = 8$$

$$= (2^{4+1} - 2^4) = (2^5 - 2^4) = 16$$

$$= (2^{5+1} - 2^5) = (2^6 - 2^5) = 32$$

$$= (2^{6+1} - 2^6) = (2^7 - 2^6) = 64$$

$$= (2^{7+1} - 2^7) = (2^8 - 2^7) = 128$$

$$= (2^{8+1} - 2^8) = (2^9 - 2^8) = 256$$

$$= 1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 + 256$$

$$= \boxed{511}$$

Question 10 30:-

What are the values of these sums,
where $S = \{1, 3, 5, 7\}$?

(a)

$$\sum_{j \in S} j$$

$$= 1 + 3 + 5 + 7$$

$$= \boxed{16}$$

(b)

$$\sum_{j \in S} j^2$$

$$= (1)^2 + (3)^2 + (5)^2 + (7)^2$$

$$= 1 + 9 + 25 + 49$$

$$= \boxed{84}$$

(c)

$$\sum_{j \in S} (1/j)$$

$$= 1/1 = 1$$

$$= \frac{1}{3} = 0.33$$

$$= \frac{1}{5} = 0.2$$

$$= \frac{1}{7} = 0.14$$

$$= 1 + 0.33 + 0.2 + 0.14$$

$$= \boxed{1.67}$$

(d)

$$\sum_{j=0}^3 1$$

$$= 1 + 1 + 1 + 1$$
$$= \boxed{4}$$

Question No 31 :-

What is the value of each of these sums of terms of a geometric progression?

(a)

$$\sum_{j=0}^8 3 \cdot 2^j$$

For 1st term:

$$3 \cdot 2^0, 3 \cdot 2^1, 3 \cdot 2^2, \dots$$

$$3 \cdot 1, 3 \cdot 2, 3 \cdot 4, \dots$$

$$\boxed{3}, 6, 12$$

For r :-

$$\frac{2 \cdot 6}{3}, \frac{12 \cdot 9}{6}$$

$$2, 2$$

Putting Values

$$S_n = a \cdot \frac{1-r^n}{1-r}$$

$$S_9 = \frac{3 \cdot (1-2^9)}{1-2}$$

$$= \frac{3 \cdot (1-512)}{1-2}$$

$$= \frac{3 \cdot (-511)}{-1} = \boxed{1533}$$

Geometric Progression :-

$$a, ar, ar^2, ar^3, ar^4, \dots$$

a = first term

r = Common Ratio Second term $\div$ first term

n = No of terms.

Formula :-

$$S_n = \frac{a \cdot (1 - r^n)}{1 - r}$$

(b)

$$\sum_{j=1}^8 2^j$$

For 1st term :-

$$2^1, 2^2, 2^3, \dots$$

For r :-

$$\frac{2^2}{2^1}, \frac{2^3}{2^2}, \dots$$

$$2, 2, 2, \dots$$

$$2, 2, 2, \dots$$

$$2, 2, 2, \dots$$

Putting Values

$$S_n = \frac{a \cdot (1 - r^n)}{1 - r}$$

$$S_8 = \frac{2 \cdot (1 - 2^8)}{1 - 2}$$

$$= \frac{2 \cdot (1 - 256)}{-1}$$

$$= \frac{2 \cdot (-255)}{-1} = \boxed{510}$$

(c)

$$\sum_{j=2}^8 (-3)^j$$

0, 1, 2, 3, 4, 5, 6, 7, 8

For 9th term:-

$$-3^2, -3^3, -3^4, \dots$$

(9), -27, 81

For r:-

$$\frac{-27}{9}, \frac{81}{-27} = -3$$

Putting Values:-

$$S_n = a \cdot \frac{(1-r^n)}{(1-r)}$$

$$S_8 = 9 \cdot \frac{(1-(-3)^8)}{(1-(-3))}$$

$$= 9 \cdot \frac{1-(-2187)}{1+3}$$

$$= 9 \cdot \frac{1+2187}{1+3}$$

$$= 9 \cdot \frac{2188}{4} = 4923$$

(d)

$$\sum_{j=0}^8 2 \cdot (-3)^j$$

0, 1, 2, 3, 4, 5, 6, 7, 8

For 9th term:-

$$2 \cdot (-3)^0, 2 \cdot (-3)^1, 2 \cdot (-3)^2, \dots$$

= 2 \cdot 1, 2 \cdot -3, 2 \cdot 9

= (2), -6, 18

For $r = -\frac{3}{2}$, $\frac{18^3}{-6}$

-3

Putting values:

$$S_n = a \cdot \frac{(1-r^n)}{(1-r)}$$

$$S_8 = \frac{2 \cdot (1 - (-3))}{(1 - (-3))}$$

$$= \frac{2 \cdot (1 - (-19683))}{1 + 3}$$

$$= \frac{2 \cdot 1 + 19683}{4}$$

$$= \frac{2 \cdot 19684}{4}$$

$$= 9842$$

Question no 32:-

Find the value of each of these sums.

(a)

$$\sum_{j=0}^8 (1 + (-1)^j)$$

Sol:-

$$= (1 + (-1)^0) = 1 + 1 = 2$$

$$- (1 + (-1)^1) = 1 + (-1) = 0$$

$$(1 + (-1)^2) = 1 + 1 = 2$$

$$(1+(-1)^3) = 1+(-1) = 0$$

$$(1+(-1)^4) = 1+1 = 2$$

$$(1+(-1)^5) = 1+(-1) = 0$$

$$(1+(-1)^6) = 1+1 = 2$$

$$(1+(-1)^7) = 1+(-1) = 0$$

$$(1+(-1)^8) = 1+1 = 2$$

$$= 2+0+2+0+2+0+2+0+2$$

$$= \boxed{10}$$

(b)

$$\sum_{j=0}^8 (3^j - 2^j)$$

Sol :-

0 to 8

$$(3^0 - 2^0) = (1 - 1) = 0$$

$$(3^1 - 2^1) = (3 - 2) = 1$$

$$(3^2 - 2^2) = (9 - 4) = 5$$

$$(3^3 - 2^3) = (27 - 8) = 19$$

$$(3^4 - 2^4) = (81 - 16) = 65$$

$$(3^5 - 2^5) = (243 - 32) = 211$$

$$(3^6 - 2^6) = (729 - 64) = 665$$

$$(3^7 - 2^7) = (2187 - 128) = 2059$$

$$(3^8 - 2^8) = (6561 - 256) = 6305$$

$$= 0 + 1 + 5 + 19 + 65 + 211 + 665 +$$

$$2059 + 6305$$

$$= \boxed{9330}$$

$$2 \times 2 \times 2 \times 2 \times 2$$

$$\begin{array}{r} 81 \\ \times 3 \\ \hline 243 \end{array}$$

$$3^0 - 2^0, 3^1 - 2^1$$

$$3^2 - 2^2$$

$$1 - 1,$$

$$3^0, 3^1, 3^2$$

$$\textcircled{1}, 3, 9$$

$$3^3, 9^3$$

(c)

$$\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j)$$

j=0

0 to 8

$$(2 \cdot 3^0 + 3 \cdot 2^0) = (2 + 3) = 5$$

$$(2 \cdot 3^1 + 3 \cdot 2^1) = (6 + 6) = 12$$

$$(2 \cdot 3^2 + 3 \cdot 2^2) = (18 + 12) = 30$$

$$(2 \cdot 3^3 + 3 \cdot 2^3) = (54 + 24) = 78$$

$$(2 \cdot 3^4 + 3 \cdot 2^4) = (162 + 48) = 210$$

$$(2 \cdot 3^5 + 3 \cdot 2^5) = (486 + 96) = 582$$

$$(2 \cdot 3^6 + 3 \cdot 2^6) = (1458 + 192) = 1650$$

$$(2 \cdot 3^7 + 3 \cdot 2^7) = (4374 + 384) = 4758$$

$$(2 \cdot 3^8 + 3 \cdot 2^8) = (13122 + 768) = 13890$$

$$= 5 + 12 + 30 + 78 + 210 + 582 + 1650 + 4758 + 13890$$

=

21215

(d)

$$\sum_{j=0}^8 (2^{j+1} - 2^j)$$

For 1st term :- 2^{j+1}

$$2^{0+1}, 2^{1+1}, 2^{2+1}$$

$$= 2^1, 2^2, 2^3$$

$$= \textcircled{2}, 4, 8$$

$$\text{For } r = \frac{2^4}{2^1}, \frac{8^2}{4^1} = \boxed{2}$$

$$2^j$$

$$2^0, 2^1, 2^2$$

$$\textcircled{1}, 2, 4$$

$$\frac{2}{1}, \frac{4}{2} = \boxed{2}$$

$$\sum_{j=0}^8 2^{j+1} = a \cdot \frac{(1-r^n)}{1-r}$$

$$= \frac{2(1-2^9)}{1-2} = 1022$$

$$\sum_{j=0}^8 2^j = \frac{a(1-r^n)}{(1-r)} = \frac{1(1-2^9)}{1-2}$$

$$= 511$$

$$= 1022 - 511 = \boxed{511}$$

Question No 33:-
Compute each of these double sums.

(a)

$$\sum_{i=1}^2 \sum_{j=1}^3 (i+j)$$

Sol :-

For $i = 1$:-

$$(1+1) + (1+2) + (1+3) = 2 + 3 + 4 = 9$$

For $i = 2$:-

$$(2+1) + (2+2) + (2+3) = 3 + 4 + 5 = 12$$

$$= 9 + 12 \\ = 21$$

(b)

$$\sum_{i=0}^2 \sum_{j=0}^3 (2i+3j)$$

For $i = 0$:-

$$(2(0)+3(0)) + (2(0)+3(1)) + (2(0)+3(2)) + (2(0)+3(3)) \\ = (0+0) + (0+3) + (0+6) + (0+9) \\ = 18$$

For $i = 1$:-

$$(2(1)+3(0)) + (2(1)+3(1)) + (2(1)+3(2)) + (2(1)+3(3)) \\ = (2+0) + (2+3) + (2+6) + (2+9) \\ = 2 + 5 + 8 + 11 \\ = 26$$

For $i = 2$:-

$$\begin{aligned} & (2(2) + 3(0)) + (2(2) + 3(1)) + (2(2) + 3(2)) + \\ & (2(2) + 3(3)) \\ &= (4 + 0) + (4 + 3) + (4 + 6) + (4 + 9) \\ &= 4 + 7 + 10 + 13 \\ &= 34 \end{aligned}$$

$$= 18 + 26 + 34$$

$$= \boxed{78}$$

(c)

$$\sum_{i=1}^3 \sum_{j=0}^2 i$$

For $i = 1$:-

$$1 + 1 + 1 = 3$$

For $i = 2$:-

$$2 + 2 + 2 = 6$$

For $i = 3$:-

$$3 + 3 + 3 = 9$$

$$3 + 6 + 9 = \boxed{18}$$

(d)

$$\sum_{i=0}^2 \sum_{j=1}^3 ij$$

For $i = 0$:-

$$(0)(1) + (0)(2) + (0)(3) = 0 + 0 + 0 = 0$$

For $i = 1$:-

$$(1)(1) + (1)(2) + (1)(3) = 1 + 2 + 3 = 6$$

For $i = 2$:-

$$(2)(1) + (2)(2) + (2)(3) = 2 + 4 + 6 = 12$$

$$(3)(1) + (3)(2) + (3)(3) = 3 + 6 + 9 = 18$$

$$= \boxed{18}$$

Question 34 :-

Compute each of these double sums.

(a)

$$\sum_{i=1}^3 \sum_{j=1}^2 (i-j)$$

For $i = 1$:-

$$= (1-1) + (1-2)$$

$$= 0 + (-1)$$

$$= -1$$

For $i = 2$:-

$$= (2-1) + (2-2)$$

$$= 1 + 0$$

$$= 1$$

For $i = 3$:-

$$= (3-1) + (3-2)$$

$$= 2 + 1$$

$$= 3$$

$$= -1 + 1 + 3$$

$$= \boxed{3}$$

(b)

$$\sum_{i=0}^3 \sum_{j=0}^2 (3i+2j)$$

For $i=0$:-

$$\begin{aligned} & (3(0)+2(0)) + (3(0)+2(1)) + (3(0)+2(2)) \\ &= (0+0) + (0+2) + (0+4) \\ &= 2+4 \\ &= 6 \end{aligned}$$

For $i=1$:-

$$\begin{aligned} & (3(1)+2(0)) + (3(1)+2(1)) + (3(1)+2(2)) \\ &= (3+0) + (3+2) + (3+4) \\ &= 3 + 5 + 7 \\ &= 15 \end{aligned}$$

For $i=2$:-

$$\begin{aligned} & (3(2)+2(0)) + (3(2)+2(1)) + (3(2)+2(2)) \\ &= (6+0) + (6+2) + (6+4) \\ &= 6 + 8 + 10 \\ &= 24 \end{aligned}$$

For $i=3$:-

$$\begin{aligned} &= (3(3)+2(0)) + (3(3)+2(1)) + (3(3)+2(2)) \\ &= (9+0) + (9+2) + (9+4) \\ &= 9 + 11 + 13 \\ &= 33 \end{aligned}$$

$$\begin{aligned} &= 33 + 24 + 15 + 6 \\ &= \boxed{78} \text{ Ans} \end{aligned}$$

(c)

$$\sum_{i=1}^3 \sum_{j=0}^2 j$$

For $j=0$:-

$$= 0 + 0 + 0 = 0$$

For $j=1$:-

$$1 + 1 + 1 = 3$$

For $j=2$:-

$$2 + 2 + 2 = 6$$

$$0 + 3 + 6 = \boxed{9}$$

(d)

$$\sum_{i=0}^2 \sum_{j=0}^3 i^2 j^3$$

For $i=0$:-

$$\begin{aligned} & ((0)^2(0)^3) + ((0)^2(1)^3) + ((0)^2(2)^3) + ((0)^2(3)^3) \\ &= (0)(0) + (0)(1) + (0)(8) + (0)(27) \\ &= 0 + 0 + 0 + 0 \\ &= 0 \end{aligned}$$

For $i=1$:-

$$\begin{aligned} &= ((1)^2(0)^3) + ((1)^2(1)^3) + ((1)^2(2)^3) + ((1)^2(3)^3) \\ &= ((1)(0)) + ((1)(1)) + ((1)(8)) + ((1)(27)) \\ &= 0 + 1 + 8 + 27 \\ &= 36 \end{aligned}$$

For $i=2$:-

$$\begin{aligned} & ((2)^2(0)^3) + ((2)^2(1)^3) + ((2)^2(2)^3) + ((2)^2(3)^3) \\ &= ((4)(0)) + ((4)(1)) + ((4)(8)) + ((4)(27)) \\ &= 0 + 4 + 32 + 108 \end{aligned}$$

$$= 144$$

$$= 0 + 144 + 36$$

$$= \boxed{180}$$

Question no 39 :-

Find $\sum_{k=100}^{200} k$

Splits the limits : $\sum_{k=1}^n k = \frac{n(n+1)}{2}$

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\therefore \sum_{k=1}^{200} k = \sum_{k=1}^{99} k + \sum_{k=100}^{200} k$$

$$\sum_{k=100}^{200} k = \sum_{k=1}^{200} k - \sum_{k=1}^{99} k$$

$$= \frac{200(200+1)}{2} - \frac{99(99+1)}{2}$$

$$= \frac{200(200+1)}{2} - \frac{99(99+1)}{2}$$

$$= 20100 - 4950$$

$$= \boxed{15150}$$

Question No 40 :-

Find $\sum_{k=99}^{200} k^3$

$$\sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$$

$$\sum_{k=99}^{200} k^3 = \sum_{k=1}^{98} k^3 + \sum_{k=99}^{200} k^3$$

$$\sum_{k=99}^{200} k^3 = \sum_{k=1}^{200} k^3 - \sum_{k=1}^{98} k^3$$

$$= \frac{n^2(n+1)^2}{4} - \frac{n^2(n+1)^2}{4}$$

$$= \frac{(200)^2(200+1)^2}{4} - \frac{(98)^2(98+1)^2}{4}$$

$$= 404010000 - 23532201$$

$$= \boxed{380,477,799}$$